



Prince Patel

M.Tech Student

Transportation Systems Engineering

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in Prince Patel

EDUCATION

M.Tech	Indian Institute of Science, Bangalore	8.7/10	2024–2026
B.Tech	Smt. S.R. Patel Engineering College, Unjha	9.59/10	2017–2021

M.TECH THESIS

- **Cyclists' Perceived Safety under Varying Traffic Conditions: Evidence from a Virtual Bicycle Simulator Study**
 - Designed a structured questionnaire and conducted controlled bicycle simulator experiments developed in Unity to evaluate cyclists' perceived safety in an Indian urban context.
 - Analysed perceived safety ratings and objective behavioural indicators using Python and R Studio, applying different statistical and econometric models.
 - Performed planning and policy-oriented analysis for urban road design, traffic management, and cyclist safety policy by assessing the role of traffic calming measures and bicycle lane provision in Indian cities.

PROFESSIONAL EXPERIENCE

- **Council on Energy, Environment and Water (CEEW)** *June 2026 – Present*
New Delhi, India **Consultant**
- **Council on Energy, Environment and Water (CEEW)** *June 2025 – July 2025*
Lucknow, Uttar Pradesh **Summer Intern**
 - Developed detailed questionnaires and carried out field surveys in Uttar Pradesh to assess the feasibility of Microbuses in India, engaging users, drivers, dealers, traffic police, PWD officials, and other government bodies.
 - Applied statistical and econometric models (linear regression, mixed regression, and latent class analysis) in Python & R to analyze Microbus system performance and deliver data-driven recommendations on service reliability and route/fleet optimization.

TECHNICAL SKILLS

- **Visualization & Documentation:** Power BI, QGIS, VISSIM, MS Office (Word, Excel, PowerPoint), LaTeX
- **Programming:** Python (NumPy, Pandas, Matplotlib, TensorFlow, Scikit-learn), R, SQL, Unity

PROJECTS

- **Multimodal Mode Choice Analysis for Last-Mile Connectivity Analysis & Visualization**
 - Analyzed large-scale metro last-mile data for Bengaluru, Delhi, and Nagpur in Python with Power BI and QGIS using statistical tests (Chi-square, ANOVA, t-test) and multivariate methods (Factor Analysis, K-Means) to capture mode-demographic patterns and travel-time variability.
 - Generated station-level accessibility and hotspot insights, revealing critical last-mile connectivity gaps that informed targeted improvements for urban transit planning.
- **Traffic Data Collection, Signal Optimization, and Multimodal Simulation Using PTV VISSIM**
 - Collected Sadashivanagar Police Station intersection traffic data (volumes, vehicle composition, speeds, queues, and signal timings) to develop and calibrate microscopic VISSIM models for delay estimation and signal optimization.
 - Modeled multimodal and first-last mile interactions (pedestrians, buses, transfers) to assess waiting, dwell, and transfer time impacts on intersection performance and user experience.
- **Travel Mode Choice Modelling in Post-Communist European Cities (Cluj-Napoca Metropolitan Area, Romania)**
 - Implemented Multinomial Logit and Nested Logit models in R with behavioural utility specification and MLE-based estimation, producing statistically significant and behaviourally consistent parameters.
 - Conducted sensitivity and elasticity analysis, using nested correlation structures to address IIA and generate robust mode-share forecasts under time, cost and network scenarios.
- **Route Network Analysis for Shortest-Path Determination in Urban Networks**
 - Implemented traffic assignment models using MSA and Frank-Wolfe algorithms.
 - Computed equilibrium flows, benchmarked algorithm performance and visualized network flow distributions using GeoJSON and Folium.
- **Smart Scientific Contributor Verification using Supervised Machine Learning**
 - Built a metadata-driven authorship verification classifier (genuine vs non-contributor) for research articles.
 - Evaluated Logistic Regression, SVM, Random Forest and **XGBoost**; performed feature engineering and hyperparameter tuning with 10-fold stratified CV (L1/L2 regularization), and **XGBoost performed best**.
- **Industrial Predictive Maintenance using Binary and Multi-Label Machine Learning**
 - Developed a feature-engineered sensor-telemetry pipeline for **binary** failure detection and **multi-label** fault diagnosis (TWF/HDF/PWF/OSF/RNF) with scaling and encoding.
 - Evaluated LogReg, SVM, RF, ANN, LightGBM and XGBoost; applied SMOTE-based stratified cross-validation and threshold tuning for imbalanced failures, **XGBoost** performed best.

- **Power Grid Cyber-Security using Unsupervised Machine Learning**
 - Engineered temporal and protocol-consistency features from IEC 61850 GOOSE traffic (inter-arrival and heartbeat deviation, sqNum–stNum transitions, rule violations) for unsupervised intrusion detection.
 - Evaluated LOF, Isolation Forest, One-Class SVM, GMM, K-Means and DBSCAN; for **binary** detection **LOF** performed best, while for **multi-class** characterization **K-Means** performed best.
- **Atliqo Bank-Credit Card Issuance and Customer Segmentation**
 - Analyzed 50,000 transaction records across 1,000 customers to understand spending behavior and credit usage patterns.
 - Performed data cleaning and exploratory analysis to identify credit card underutilization among the 18–25 age group. Designed a targeted credit card strategy and validated its impact through a 2-month A/B experiment.

POSITIONS OF RESPONSIBILITY

- **Placement Representative**, OCCAP, IISc Bangalore *Nov 2024 - Ongoing*
- **Placement Representative**, Smt. SR Patel Engineering College *Decemeber 2020 - February 2021*

AWARDS AND HONOURS AND ACHIEVEMENT

- GATE Achievements: All India Rank 148 (Environmental Science, 2024) and All India Rank 538 (Civil Engineering, 2024).
- Academic Distinctions: Secured Department Rank 2 in B.Tech and BIS Qualified (2021)